Switching: Packet & Circuit Switching

Lecture 12 — Summarised Notes

1. Introduction to Switching

A network is a set of connected devices. When multiple devices need to communicate, a naive solution like a full mesh topology (point-to-point links between every pair) or star topology becomes impractical and wasteful for large networks — too many links are needed and most remain idle. Multipoint topologies such as buses also fail beyond a certain scale. The better solution is a **switched network**, which consists of a series of interlinked nodes called **switches**. Switches create temporary connections between devices linked to them. Some nodes connect to end systems (computers, phones) while others handle only routing.

1.1 Three Methods of Switching

- **Circuit Switching** — operates at the Physical Layer.
- **Packet Switching** — operates at the Data-Link or Network Layer; subdivided into **Virtual-Circuit** and **Datagram** approaches.
- **Message Switching** — operates at the Application Layer (e.g., e-mail).

1.2 Switching at TCP/IP Layers

Layer	Switching Type	Notes
Physical	Circuit Switching	No packets; signals travel one path or another.
Data-Link	Packet Switching	Frames/cells; virtual-circuit approach used.
Network	Packet Switching	Virtual-circuit or datagram approach.
Application	Message Switching	Messages exchanged (e.g., e-mail).

---

2. Circuit-Switched Networks

A circuit-switched network is made of a set of switches connected by physical links, where each link is divided into *n* channels using **FDM (Frequency Division Multiplexing)** or **TDM (Time Division Multiplexing)**. A connection between two stations forms a dedicated path made of one or more links, and each connection uses only one dedicated channel on each link. Resources such as channels, switch buffers, processing time, and I/O ports are reserved for the entire duration of the call and released only at teardown. Data flow is continuous (not packetised) and there is no addressing during data transfer — switches route based on FDM band or TDM time slot.

2.1 Three Phases of Circuit Switching

- **Setup Phase:** A dedicated circuit (channel combination) is established across switches. Source sends a setup request; each switch reserves a channel; destination acknowledges; setup is complete only after A receives the acknowledgment.
- **Data-Transfer Phase:** Data flows continuously along the dedicated path. No waiting at switches; no re-addressing needed.
- **Teardown Phase:** One party signals disconnection; all switches release the reserved resources.

Key Rule: *In circuit switching, resources must be reserved during the setup phase and remain dedicated for the entire data-transfer duration until teardown.*

2.2 Efficiency

Circuit-switched networks are the least efficient of the three types because resources are locked for the full connection duration even during silence. In telephone networks this is acceptable; in computer networks it is wasteful when links are idle for long periods.

2.3 Delay

Despite low efficiency, delay is **minimal** during data transfer because resources are pre-allocated and there is no queuing at switches. Total delay = Setup delay (request propagation + transfer + acknowledgment propagation + transfer) + Data-transfer delay (propagation + transmission time) + Teardown delay.

2.4 Advantages

- Guaranteed, constant bandwidth for the entire session.
- Very low and predictable delay during data transfer — no queuing at intermediate nodes.
- Data arrives in order and without gaps (continuous stream).
- Ideal for real-time, delay-sensitive applications such as voice calls.

2.5 Disadvantages

- Inefficient use of bandwidth — channel is reserved even during silence or idle periods.
- Call setup delay before any data can be sent.
- No flexibility — fixed bandwidth even if the application needs less.
- Resources wasted if the connection drops or is rarely used (common in computer networks).

---

3. Packet Switching — Introduction

In packet switching, a message is divided into **packets** of fixed or variable size determined by the network and governing protocol. Unlike circuit switching, **there is no resource reservation** for individual packets — resources are allocated on demand (first-come, first-served). When a switch receives a packet it must wait in a queue if other packets are being processed. Packet switching has two variants: **Datagram Networks** and **Virtual-Circuit Networks**.

Key Rule: *In a packet-switched network there is no resource reservation; resources are allocated on demand.*

---

4. Datagram Networks

In a datagram network, each packet (called a **datagram**) is treated independently of all others, even if it belongs to the same multi-packet message. Datagram switching is performed at the **Network Layer**. The switches in a datagram network are traditionally called **routers**. Because every packet is routed independently, packets of the same message may travel *different paths* and arrive out of order or with variable delays. Upper-layer protocols are responsible for reordering or requesting retransmission of lost packets.

4.1 Routing Table

Because there are no setup or teardown phases, each router maintains a dynamic **routing table** (updated periodically) indexed by destination address. When a packet arrives, its destination address is looked up and the packet is forwarded out the corresponding port. The destination address in a datagram's header **remains unchanged** throughout the entire journey.

4.2 Connectionless Nature

Datagram networks are **connectionless** — routers maintain no state about individual connections. Every packet is treated the same regardless of source or destination. There are no setup or teardown phases.

4.3 Efficiency

Efficiency is **better than circuit switching** because resources are only used when packets are actually present. Idle time between packets can be used by other sources.

4.4 Delay Formula

For a packet travelling through two switches:

Total delay = $3T + 3\tau + w_1 + w_2$

where T = transmission time per link, τ = propagation delay per link, w = waiting time at each switch. Delay is non-uniform across packets of the same message.

4.5 Advantages

- No setup or teardown delay — packets can be sent immediately.
- Efficient bandwidth use — resources allocated only when data is present.
- Robust — if a router fails, packets can be rerouted dynamically.
- No wasted bandwidth during idle periods.
- Suitable for bursty, non-real-time data traffic.

4.6 Disadvantages

- Packets may arrive out of order; upper layers must resequence.
- Variable delay (jitter) — unsuitable for real-time audio/video without extra mechanisms.
- Packets can be lost or dropped due to congestion (no guaranteed delivery).
- Each packet carries a full destination address, adding overhead.
- Routing table lookups add processing overhead at every hop.

---

5. Virtual-Circuit Networks

A virtual-circuit network is a **hybrid** between circuit switching and datagram networking. It operates at the **Data-Link Layer**. Like circuit switching, it has setup and teardown phases and all packets of a session follow the same path. Like datagram switching, data is packetised and resources may be allocated on demand. The switches in a virtual-circuit network maintain a **switching table** that maps incoming (port, VCI) pairs to outgoing (port, VCI) pairs.

5.1 Addressing

- **Global Address:** A globally unique address (like a phone number or IP address) used only during the setup phase to establish the virtual circuit.
- **Virtual-Circuit Identifier (VCI) / Label:** A small, locally-scoped number used during data transfer. It identifies the channel between two adjacent switches. The VCI *changes at every switch* as the frame passes through — this is called label swapping.

5.2 Three Phases

A. Setup Phase

Source A sends a *setup request frame* carrying the global destination address to the first switch. Each switch along the path: (1) consults its routing table to determine the outgoing port, (2) picks an available incoming VCI, (3) adds an entry with incoming port, incoming VCI, and outgoing port (outgoing VCI is filled during acknowledgment), and (4) forwards the setup frame. The destination B receives the setup frame and assigns an incoming VCI (e.g., 77) for frames from A.

B. Acknowledgment (completing setup)

B sends an acknowledgment back toward A. Each switch on the return path uses the VCI from the next switch as its outgoing VCI, completing its table entry. Once A receives the acknowledgment, the virtual circuit is fully established.

C. Data-Transfer Phase

A sends data frames with a VCI. Each switch looks up the (incoming port, incoming VCI), replaces the VCI with the outgoing VCI from its table, and forwards the frame out the outgoing port. All frames follow the **same path**. The VCI changes hop-by-hop but the physical path is fixed.

D. Teardown Phase

A sends a special *teardown request* frame after all data has been sent. B responds with a *teardown confirmation*. Each switch deletes the corresponding entry from its switching table, releasing the resources.

Key Rule: *In virtual-circuit switching, all packets belonging to the same source and destination travel the same path, but packets may arrive with different delays if resource allocation is on demand.*

5.3 Delay

There is a one-time setup delay and a one-time teardown delay. If resources are allocated during setup, there is **no waiting time** for individual packets at switches (similar to circuit switching). If resources are allocated on demand, waiting is possible.
Total delay = 3T + 3τ + setup delay + teardown delay (for a path through two switches).

5.4 Advantages

- All packets follow the same path — arrive in order (no resequencing needed).
- Low per-packet overhead — VCI is a small local number, not a full global address.
- Predictable QoS possible — resources can be reserved at setup time.
- Better suited for real-time or delay-sensitive traffic than datagram networks.
- Combines benefits of both circuit and datagram switching.

5.5 Disadvantages

- Setup and teardown overhead before data transfer can begin.
- If a switch on the virtual circuit fails, the entire connection is broken.
- More complex switch logic — must maintain per-connection state tables.
- Not as robust as datagram networks (no dynamic rerouting per packet).
- Global addresses still needed for setup, adding initial complexity.

---

6. Comparison of Switching Methods

Feature	Circuit Switching	Datagram (Packet)	Virtual-Circuit (Packet)
Layer	Physical	Network	Data-Link
Resource Allocation	Reserved at setup	On demand	Setup or on demand
Path	Fixed dedicated path	Each packet independent	Fixed for all packets
Addressing in data	None (FDM/TDM slot)	Full dest. address	Local VCI (label)
Setup/Teardown	Required	Not required	Required
Packet ordering	In-order (continuous)	May be out of order	In-order
Efficiency	Low (idle waste)	High	Medium-High
Delay	Low (after setup)	Variable (queuing)	Low (after setup)
Robustness	Low (single path)	High (dynamic routing)	Medium

Typical use	Voice calls (PSTN)	Internet (IP)	ATM, Frame Relay

---

7. Quick Revision — Key Definitions

Switched Network: A network of interlinked switches that create temporary connections between end systems, avoiding the cost of dedicated point-to-point links.

Circuit Switching: A switching method where a dedicated, end-to-end path of channels is established before data transfer and held until explicitly released.

Packet Switching: A switching method where the message is broken into packets; each packet is routed independently (datagram) or along a pre-established virtual path (virtual-circuit), with resources allocated on demand.

Datagram: An independent packet in a connectionless network that carries the full destination address and may be routed independently of other packets in the same message.

Virtual Circuit: A logical, pre-established path between source and destination in which all packets follow the same route and each link uses a locally-scoped VCI instead of a full global address.

VCI (Virtual-Circuit Identifier): A small, locally-significant label attached to a frame in a virtual-circuit network. It is swapped (replaced) at each switch hop.

FDM: Frequency Division Multiplexing — divides a link's bandwidth into frequency bands, each assigned to one channel.

TDM: Time Division Multiplexing — divides a link's time into slots, each assigned to one channel.

Routing Table: A dynamically maintained table in a datagram router that maps destination addresses to output ports.

Switching Table: A per-connection table in a virtual-circuit switch mapping (incoming port, incoming VCI) to (outgoing port, outgoing VCI).

These notes summarise Lecture 12 on Switching. All topics from the original slides are covered including switched networks, TCP/IP layer mapping, circuit switching (phases, efficiency, delay), packet switching introduction, datagram networks (routing table, destination addressing, efficiency, delay), and virtual-circuit networks (addressing, VCI, all three phases, delay).